Encapsulation

Encapsulation is the process of wrapping up of data and member function in a single unit called class and restricting direct access to some of the object details. It helps to protect data and secure controlled access.

→ Key features -

1.) Data hiding - Prevents direct access to the internal state of an object, protecting data from accidental modification.

2.) Control access - Allows controlled access through getter and setter methods.

3.) Enhance security - Restricts unauthorized access.

→ C++ program for Encapsulation

```
#include<iostream>
using namespace std;
class student
{
    private:

    int age;    // private data member
    int roll;
    string name;
```

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```
public :

void setage(int a) // setter methods to set values
{
    age = a;
}

int getage() // getter methods to get values
{
    return age;
}

void setroll(int r)
{
    roll = r;
}

int getroll()
{
    return roll;
}

void setname(string n[redacted])
{
    name = n;
}

string getname()
{
```

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```
    return name;
  }
};
int main()
  {
    Student s;
    s.setage(20);
    s.setroll(12355);
    s.setname("akash kumar");
    cout<<"age = "<<s.getage()<<endl;
    cout<<"roll= "<<s.getroll()<<endl;
    cout<<" name= "<<s.getname();
    return 0;
  }
```

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